
Reproducing “Bilinear MLPs Enable Weight-Based Mechanistic Interpretability”

Abstract

1 Introduction

2 Scope of Reproducibility

We reproduce Section 4 (Vision) of Pearce et al. (2024), which demonstrates that bilinear MLPs enable weight-based mechanistic interpretability through eigendecomposition of the interaction tensor.

2.1 Claims to Verify

The original paper makes the following testable claims regarding the vision experiments:

- **Claim 1:** Regularization (input noise + weight decay) induces low-rank structure in the bilinear interaction tensor, as measured by effective rank.
- **Claim 2:** The eigenvectors of the regularized model’s interaction tensor correspond to interpretable digit-like patterns.
- **Claim 3:** Without regularization, the eigenspectrum is flat (high effective rank) and eigenvectors show noise-like overfitting patterns.

2.2 Paper Context

3 Methodology

3.1 Bilinear Layer

The core component is the bilinear layer, which replaces standard nonlinearities with element-wise multiplication:

$$\mathbf{y} = (\mathbf{W}_l \mathbf{x}) \odot (\mathbf{W}_r \mathbf{x}) \quad (1)$$

where $\mathbf{W}_l, \mathbf{W}_r \in \mathbb{R}^{d_{out} \times d_{in}}$ are learnable weight matrices and $\odot$ denotes the Hadamard (element-wise) product.

This can be expressed as a third-order interaction tensor:

$$y_i = \sum_{j,k} B_{ijk} x_j x_k \quad (2)$$

where $B_{ijk} = (W_l)_{ij}(W_r)_{ik}$.

3.2 Eigendecomposition for Interpretability

The interaction tensor $\mathbf{B}$ can be reshaped and decomposed via eigendecomposition to reveal interpretable structure. For each output dimension i , the matrix $B_i \in \mathbb{R}^{d_{in} \times d_{in}}$ admits:

$$B_i = \sum_r \lambda_r^{(i)} \mathbf{v}_r^{(i)} (\mathbf{v}_r^{(i)})^\top \quad (3)$$

3.3 Effective Rank

We measure the interpretability of the learned representation using effective rank (Roy & Vetterli, 2007):

$$\text{EffRank}(\mathbf{B}) = \exp \left(- \sum_i p_i \log p_i \right), \quad p_i = \frac{|\lambda_i|}{\sum_j |\lambda_j|} \quad (4)$$

Lower effective rank indicates more concentrated eigenspectrum, corresponding to more interpretable structure.

3.4 Extension: CP-Decomposition Layer

As an extension, we implement a CP-decomposed bilinear layer with explicit rank control:

$$B_{ijk} = \sum_{r=1}^R \lambda_r A_{ir} B_{jr} C_{kr} \quad (5)$$

where R is the chosen rank. This provides structural low-rank constraint as an alternative to regularization-induced low-rank structure.

3.5 Datasets

3.6 Hyperparameters

3.7 Computational Requirements

4 Reproduction Results

4.1 Eigenspectrum Analysis

4.2 Eigenvector Visualization

4.3 Ablation Study

4.4 Quantitative Results

5 Extension Results

5.1 Robustness Analysis

5.2 Transfer to EMNIST

5.3 CP-Decomposition Analysis

5.4 Structure vs. Regularization

6 Discussion

6.1 What was Easy

6.2 What was Difficult

6.3 Environmental Impact

6.4 Connection to FACT-AI Topics

6.5 Communication with Original Authors

7 Conclusion

Broader Impact Statement

This work contributes to the field of mechanistic interpretability, which aims to make neural networks more transparent and understandable. Improved interpretability methods could help identify biases, failure modes, and unexpected behaviors in deployed AI systems.

Author Contributions

Acknowledgments

References

Tim Pearce, Tuomas Oikarinen, and Tsui-Wei Weng. Bilinear mlps enable weight-based mechanistic interpretability. *arXiv preprint arXiv:2410.08417*, 2024.

Olivier Roy and Martin Vetterli. The effective rank: A measure of effective dimensionality. In *European Signal Processing Conference (EUSIPCO)*, pp. 606–610, 2007.

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- A Additional Figures**
 - B Hyperparameter Details**
 - C CO2 Tracking Details**